

HOFU AB, Japan

WMO: 477880

Lat: 34.034N

Lon: 131.546E

Elev: 17

StdP: 14.69

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	28.3	30.1	17.5	13.3	37.1	19.6	14.8	38.1	24.7	41.0	22.2	41.1	6.1	10	0.435

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.3	91.3	79.2	89.5	78.9	87.6	78.6	81.1	87.0	80.1	85.9	79.4	84.8	10.0	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.2	151.7	84.1	78.8	149.6	83.7	77.4	142.5	82.6	44.9	87.5	43.9	86.7	43.0	84.6	87.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
22.0	18.8	16.5	DB	24.2	94.8	3.1	2.1	22.0	96.3	20.1	97.5	18.4	98.7	16.1	100.3
			WB	22.4	82.7	2.7	1.5	20.4	83.8	18.8	84.7	17.3	85.6	15.3	86.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	60.7	41.0	42.4	48.1	56.9	65.5	72.3	79.6	81.7	75.1	65.0	54.6	45.0
	DBStd	14.81	4.55	5.25	5.54	5.52	3.94	3.51	3.66	3.07	4.67	5.24	5.99	5.45
	HDD50	811	282	220	104	11	0	0	0	0	0	0	22	172
	HDD65	3190	745	632	525	247	42	1	0	0	0	64	315	619
	CDD50	4713	3	8	44	217	480	668	918	981	752	464	160	18
	CDD65	1615	0	0	0	3	57	219	453	516	302	63	2	0
	CDH74	13369	0	0	0	18	257	964	4145	5528	2217	240	0	0
	CDH80	4018	0	0	0	0	21	120	1334	2026	506	11	0	0
	WSAvg	7.3	7.2	7.3	8.0	8.1	7.9	7.0	7.4	7.5	7.0	6.6	6.6	7.1
	PrecAvg	68.5	3.2	3.2	5.0	5.7	6.8	9.4	11.0	6.8	6.8	3.8	3.7	3.1
(15) Precipitation	PrecMax	96.4	6.9	4.9	9.4	11.4	14.1	17.8	24.5	19.8	14.5	8.9	8.3	6.1
	PrecMin	46.9	0.9	1.1	2.4	1.7	2.1	1.9	2.0	1.4	2.1	0.5	1.1	0.5
	PrecStd	12.4	1.4	1.1	1.9	2.4	2.8	4.1	6.7	4.1	3.2	2.3	1.9	1.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.8	62.6	68.0	76.7	82.5	86.1	93.4	93.7	89.5	81.0	72.0	63.0
		MCWB	53.3	57.0	56.1	63.3	64.3	76.4	79.7	79.7	78.5	72.0	64.0	56.3
	2%	DB	55.2	58.8	64.4	72.0	79.0	82.8	91.1	91.7	86.4	78.9	69.9	60.6
		MCWB	49.3	53.0	54.0	60.0	63.9	72.9	79.2	79.3	76.9	70.3	62.8	54.1
	5%	DB	52.2	55.5	61.2	69.8	76.9	80.9	89.2	89.9	84.6	76.9	67.8	57.5
		MCWB	46.2	49.3	53.5	59.2	64.0	71.8	79.0	78.9	75.9	68.5	61.0	51.6
	10%	DB	50.2	52.2	59.0	67.6	74.9	79.0	86.3	88.1	82.7	74.9	65.8	55.4
		MCWB	44.4	46.5	52.3	58.3	63.6	71.0	78.2	78.9	74.6	66.5	59.6	49.6
	0.4%	WB	55.5	58.6	60.5	68.2	71.0	78.8	82.1	83.4	80.7	75.3	68.4	60.0
		MCDB	57.3	61.3	63.7	70.9	75.2	83.4	88.6	89.6	85.4	78.3	70.5	61.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	51.1	54.5	57.8	64.1	69.2	76.5	80.8	81.7	79.1	72.8	65.2	55.6
		MCDB	53.3	57.6	61.3	68.3	73.4	80.2	87.1	87.4	83.7	76.9	68.0	58.8
	5%	WB	47.8	51.1	55.6	62.0	67.4	74.8	79.9	80.6	78.0	70.2	62.3	52.9
		MCDB	51.1	54.1	59.9	66.9	72.6	78.3	85.9	86.6	82.6	74.8	66.1	56.7
	10%	WB	45.0	48.1	53.0	60.1	65.8	73.5	79.1	79.8	76.8	67.8	60.0	50.4
		MCDB	49.0	51.8	58.0	65.2	71.4	76.8	84.7	85.4	81.4	72.7	64.5	54.4
(35) Mean Daily Temperature Range	5% DB	MDBR	15.3	16.0	17.2	18.0	16.8	12.2	11.2	12.3	13.4	16.2	16.6	15.8
		MCDBR	17.8	18.9	20.5	21.6	21.6	15.5	15.2	15.0	13.9	16.4	16.7	17.6
		MCWBR	12.8	13.7	13.4	12.3	11.2	8.1	6.8	6.5	7.1	9.5	11.0	12.7
		MCDBR	15.3	15.6	15.8	15.2	14.1	10.2	12.7	12.3	11.7	13.2	14.0	15.3
	5% WB	MCWBR	13.1	13.6	12.7	10.4	9.4	6.3	6.3	6.0	6.5	9.2	11.0	12.8
(40) Clear-Sky Solar Irradiance	taub		0.438	0.482	0.550	0.577	0.578	0.592	0.592	0.553	0.514	0.467	0.441	0.429
	taud		2.029	1.947	1.798	1.787	1.842	1.891	1.936	2.039	2.098	2.124	2.108	2.063
	Ebn at Noon		233	236	231	233	234	231	230	236	239	239	231	226
	Edh at Noon		43	52	65	69	66	63	60	53	48	43	40	40
(44) All-Sky Solar Radiation	RadAvg		667	899	1238	1518	1663	1436	1540	1617	1279	1073	790	615
	RadStd		59	81	90	134	184	155	227	172	132	111	66	41

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (30 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page